

# MARK APOLLO V. CAYETANO

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Province

## PROFESSIONAL SUMMARY

Computer Engineering graduate with a strong foundation in hardware-software integration and full-stack development. Proven ability to build responsive web architectures and automated systems using Node.js and MongoDB. Seeking to bring advanced analytical thinking and logical problem-solving skills to a forward-thinking development team.

## EDUCATION

### Bachelor of Science in Computer Engineering

Southern Luzon State University (SLSU Main Campus) | September 2022 – June 2026

- Certifications: Civil Service Professional Level Eligibility (March 2026)

## TECHNICAL SKILLS

- Languages: C/C++, Python (automation), JavaScript, Java
- Front-End Development: Angular (v20), TypeScript, HTML, CSS responsive design
- Back-End & Database: Node.js, RESTful APIs, MongoDB, data modeling, real-time handling
- Hardware & IoT: Raspberry Pi, Arduino, UART, hardware-software integration
- Systems & Tools: Kiosk mode, autostart scripting, POS logic, EasyEDA, Git/GitHub, EditorConfig
- Foundations: Computer architecture, networking, problem-solving

## RELEVANT TECHNICAL PROJECTS

### Operational Management System with Playtime Notification Alert for Kidzooona Indoor Playground

Undergraduate Thesis (2026)

- Developed a self-service kiosk to automate playground operations, including registration, sock dispensing, payments, and digital receipts.
- Built a kiosk-mode system using Node.js and Raspberry Pi, integrating a bill acceptor via serial communication.
- Designed a MongoDB database to handle real-time customer data and transaction records.

### CO<sub>2</sub> Sink: A Mini Closed-Loop Algae Carbon Capture Device for Sequestration and Storage

Capstone Project (2022)

- Tested multiple households to compare CO<sub>2</sub> levels (PPM) with and without the algae-based device.
- Applied a one-tailed paired t-test to assess pretest vs. posttest differences, confirming significant CO<sub>2</sub> reduction.
- Measured light levels (lumens) and time (military hours) to identify optimal conditions for algae efficiency.

## PROFESSIONAL EXPERIENCE

### Front-End Web Developer Intern (OJT)

Picto Quezon | 2025

- Built the Picto Task Reminder using Angular 20.
- Created dynamic, responsive interfaces with TypeScript for efficient state management.
- Managed accurate reminder scheduling using the Luxon library.

## ACADEMIC AFFILIATIONS

- Member, Institute of Computer Engineers of the Philippines (Region IV-A) | 2024 – 2025
- Member, SLSU Society of Computer Engineering Students (SCoPES) | 2022 – 2026